EXPONENTIALLY MANY HAMILTONIAN SUBSETS FROM THE CRUX

ANONYMOUS

ABSTRACT. A set of vertices is Hamiltonian if it induces a subgraph containing a Hamiltonian cycle. Cambie, Gao and Liu proved that, for an n-vertex graph G of sufficiently large average degree and $t = c_{1/5}(G)$, the number $h(G)$ of Hamiltonian subsets satisfies $h(G) \geq n2^{\Omega(t/\log^{16} t)}$. They asked whether the polylogarithmic loss is necessary. We prove that it is not: there are absolute constants $\kappa, c, d_0 > 0$ such that $h(G) \geq \kappa n 2^{ct}$ whenever $\overline{d}(G) \geq d_0$. The new ingredient is a rooted counting lemma obtained by combining a long-cycle theorem for the crux with random restrictions and double counting; when the relevant expander is very large, the wheel construction of Cambie, Gao and Liu already gives the required bound.

1. INTRODUCTION

A set $A \subseteq V(G)$ is a *Hamiltonian subset* of a graph G if $G[A]$ contains a Hamiltonian cycle. Let $h(G)$ denote the number of Hamiltonian subsets of G. Komlós conjectured that, among graphs of minimum degree at least d, the complete graph K_{d+1} minimizes $h(G)$. Kim, Liu, Sharifzadeh and Staden proved the conjecture for all sufficiently large d; they also established a stronger stability theorem under an average-degree assumption [3].

For a nonempty graph G, write $\overline{d}(G) = 2e(G)/|V(G)|$. Given $0 < \alpha < 1$, its α-*crux* is

$$c_\alpha(G) = \min\{|V(H)| : H \subseteq G \text{ and } \overline{d}(H) \geq \alpha \overline{d}(G)\}.$$

Thus $c_\alpha(G)$ is the minimum number of vertices needed to retain an α-fraction of the average degree. The parameter was introduced by Haslegrave, Hu, Kim, Liu, Luan and Wang [2].

Cambie, Gao and Liu proved that there are absolute constants $B, d_0 > 0$ such that every n-vertex graph G with $\overline{d}(G) \geq d_0$ and $t = c_{1/5}(G)$ satisfies

$$h(G) \geq \frac{n}{B} 2^{\beta t/\log^{16} t}$$

for an absolute constant $\beta > 0$ [1, Theorem 1.1]. They asked whether the factor $\log^{16} t$ in the exponent is necessary. Our main result removes it.

Theorem 1.1. *There are absolute constants $\kappa, c, d_0 > 0$ such that every n-vertex graph G with $\overline{d}(G) \geq d_0$ and $t = c_{1/5}(G)$ satisfies*

$$h(G) \geq \kappa n 2^{ct}.$$

We retain the global counting scheme of Cambie, Gao and Liu and replace its rooted input. Let H be a dense sublinear expander extracted from an auxiliary graph, set $N = |V(H)|$, and let $s = c_{9/10}(H)$. The extraction guarantees $s \geq t$. If $N > 2^{\eta t}$, the wheel constructed in [1] has size $\Omega(N/\log^{16} N) = \omega(t)$ and already yields a vertex contained in $2^{\Omega(t)}$ Hamiltonian subsets.

The new argument treats $N \leq 2^{\eta t}$. A positive proportion of the induced subgraphs obtained by retaining $0.99N$ vertices have average degree close to that of H, and each such restriction has $(19/20)$-crux at least s. The long-cycle theorem of Haslegrave et al. [2] then supplies a cycle on $\Omega(s)$ vertices in every good restriction. Double counting gives $2^{\Omega(s)}$ distinct Hamiltonian subsets, each of order $\Omega(s)$. Averaging vertex–subset incidences and using $N \leq 2^{\eta t}$ produces a common vertex contained in $2^{\Omega(t)}$ of them.

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2. Preliminaries

Since every r-vertex graph has average degree at most $r-1$, we shall use repeatedly the elementary inequality

$$c_\alpha(G) > \alpha\overline{d}(G). \tag{1}$$

For $\varepsilon, k > 0$, define

$$\rho_{\varepsilon,k}(x) = \begin{cases} 0, & x < k/5, \\ \varepsilon/\log^2(15x/k), & x \geq k/5. \end{cases}$$

A graph H is an (ε, k)-*expander* if every $X \subseteq V(H)$ with $k/2 \leq |X| \leq v(H)/2$ satisfies $|N_H(X)| \geq \rho_{\varepsilon,k}(|X|)|X|$.

We use the following four results. The first is [1, Lemma 2.2] with $C = 40$, $k = 15$ and $\varepsilon_0 = \log 3/800$.

Lemma 2.1. *Every graph F contains a subgraph H that is an $(\varepsilon_0, 15)$-expander and satisfies $\overline{d}(H) \geq 19\overline{d}(F)/20$.*

An ℓ-wheel is the graph obtained from a cycle by replacing ℓ of its edges with ℓ cycles, as in [1, Definition 2.6]. The following is the consequence of [1, Lemma 2.7 and the proof of Theorem 1.2] that we use.

Lemma 2.2. *There are constants $N_{\mathrm{w}} \in \mathbb{N}$ and $\rho > 0$ such that every $(\varepsilon_0, 15)$-expander H of order $N \geq N_{\mathrm{w}}$ contains an ℓ-wheel W with $\ell \geq \rho N/\log^{16} N$, and every vertex of W belongs to at least $2^{\ell-1}$ distinct Hamiltonian subsets of H. One may take $\rho = 2(\varepsilon_0/20)^{16}$.*

For completeness, each replacement cycle has two internally disjoint paths between its attachment vertices. Choosing one path in each replacement cycle gives 2^ℓ cycles with distinct vertex sets. An attachment vertex occurs in all choices, while every other wheel vertex occurs in exactly half; hence each wheel vertex lies in at least $2^{\ell-1}$ of the resulting Hamiltonian subsets.

Theorem 2.3 (Haslegrave–Hu–Kim–Liu–Luan–Wang)**.** *For every graph J and every $0 < \alpha < 1$, there is a cycle of length at least $(1-\alpha)c_\alpha(J)/16000$, where a single edge is permitted to count as a cycle of length one.*

This is [2, Theorem 1.2]. In every application below, the stated lower bound is at least three, so the exceptional convention is irrelevant.

Proposition 2.4. *If $0 < \alpha < \alpha' < 1$, then every graph J satisfies*

$$c_\alpha(J) \leq \lceil \frac{\alpha}{\alpha'}(c_{\alpha'}(J) - 1) + 1 \rceil.$$

This is [1, Proposition 2.9].

3. Random restrictions

The next lemma is the new counting input. It does not require expansion.

Lemma 3.1. *There are absolute constants $\mu > 0$ and $s_0 \in \mathbb{N}$ such that the following holds. Let H be an N-vertex graph and set $s = c_{9/10}(H)$. If $s \geq s_0$, then H has a family $\mathcal{A}$ of at least $2^{\mu s}$ distinct Hamiltonian subsets, each of order at least $\lfloor s/320000 \rfloor$.*

Consequently, some vertex of H belongs to at least $2^{\mu s}/N$ members of $\mathcal{A}$. If H is bipartite, the vertex may be required to lie in either prescribed side of a fixed bipartition.

Proof. Set $p = 99/100$, $\tau = 24/25$, $\alpha_1 = 19/20$ and $\lambda = (1-\alpha_1)/16000 = 1/320000$. Let $q = \lfloor pN \rfloor$, choose a q-set $R \subseteq V(H)$ uniformly at random, and put $m = e(H)$ and $X = e(H[R])$. Since s is large, $m > 0$. Moreover,

$$\mathbb{E}X = m\frac{\binom{q}{2}}{\binom{N}{2}}.$$

Choose s_0 large enough that $N \geq s \geq s_0$ implies $\binom{q}{2}/\binom{N}{2} \geq 97/100$. If $\pi = \mathbb{P}(X \geq \tau m)$, then

$$\frac{97}{100}m \leq \mathbb{E}X \leq \tau m(1-\pi) + m\pi,$$

so $\pi \geq 1/4$. Call such an R good.

Fix a good R. Since $q \leq pN$,

$$\overline{d}(H[R]) = \frac{2X}{q} \geq \frac{\tau}{p}\overline{d}(H).$$

Now $\alpha_1\tau/p = 152/165 > 9/10$. Hence every $J \subseteq H[R]$ with $\overline{d}(J) \geq \alpha_1\overline{d}(H[R])$ also satisfies $\overline{d}(J) > 9\overline{d}(H)/10$, and therefore $v(J) \geq s$. Thus

$$c_{19/20}(H[R]) \geq s. \tag{2}$$

In particular, $q \geq s$. By theorem 2.3, $H[R]$ contains a cycle of length at least λs. Put $L = \lfloor \lambda s \rfloor$ and increase s_0, if necessary, so that $L \geq 3$.

For each good R, choose the vertex set A_R of one such cycle, and let $\mathcal{A}$ be the collection of distinct sets obtained. Fix $A \in \mathcal{A}$ and write $a = |A| \geq L$. At most $\binom{N-a}{q-a} \leq \binom{N-L}{q-L}$ choices of R contain A; the inequality follows, for example, from

$$\frac{\binom{N-a-1}{q-a-1}}{\binom{N-a}{q-a}} = \frac{q-a}{N-a} \leq 1.$$

Double counting the pairs (R, A_R) yields

$$|\mathcal{A}| \geq \frac{1}{4}\frac{\binom{N}{q}}{\binom{N-L}{q-L}} = \frac{1}{4}\prod_{i=0}^{L-1}\frac{N-i}{q-i} \geq \frac{1}{4}p^{-L},$$

because $(N-i)/(q-i) \geq N/q \geq 1/p$ for $0 \leq i < L \leq q$. Let $a_0 = \log_2(1/p)$ and $\mu = \lambda a_0/4$. After a further increase of s_0, we have $L \geq \lambda s/2$ and $-2 + a_0 L \geq \mu s$, whence $|\mathcal{A}| \geq 2^{\mu s}$.

Each member of $\mathcal{A}$ has at least $L \geq 3$ vertices. Averaging the at least $L|\mathcal{A}|$ vertex–set incidences over the N vertices gives a vertex contained in at least $2^{\mu s}/N$ members. If H is bipartite, every cycle meets the two sides equally. Thus each prescribed side contributes at least one incidence per member of $\mathcal{A}$, and averaging within that side gives the same bound. □

4. A rooted exponential bound

Lemma 4.1. *There are absolute constants $c, d_1 > 0$ such that the following holds. Let G be a graph with $\overline{d}(G) = d \geq d_1$ and $t = c_{1/5}(G)$. If $F \subseteq G$ satisfies $\overline{d}(F) \geq d/4$, then some vertex of F belongs to at least 2^{ct} distinct Hamiltonian subsets of F.*

If F is bipartite, the vertex may be required to lie in either prescribed side of a fixed bipartition.

Proof. Let μ, s_0 be given by theorem 3.1, and set $\eta = \mu/4$ and $c = \mu/16$. Apply theorem 2.1 to F, obtaining an $(\varepsilon_0, 15)$-expander $H \subseteq F$ with $\overline{d}(H) \geq 19d/80$. Write $N = v(H)$ and $s = c_{9/10}(H)$. Since $9\overline{d}(H)/10 \geq 171d/800 > d/5$, every $(9/10)$-crux of H is a $(1/5)$-crux candidate in G. Hence

$$s \geq t. \tag{3}$$

Also, (1) gives $t > d/5$.

Suppose first that $N \leq 2^{\eta t}$. Once d_1 is large enough that $s \geq t \geq s_0$, theorem 3.1 gives a vertex of H contained in at least

$$\frac{2^{\mu s}}{N} \geq 2^{(\mu-\eta)t} \geq 2^{ct}$$

Hamiltonian subsets of H. Its bipartite side-specific conclusion applies when required.

Now suppose that $N > 2^{\eta t}$. Let ρ, N_{w} be given by theorem 2.2. The function $x/\log^{16} x$ is increasing for $x > e^{16}$, and

$$\rho\frac{2^{\eta t}}{(\eta t \log 2)^{16}} - 1 \geq ct$$

for all sufficiently large t. Increase d_1 so that $t > d/5$ exceeds this threshold and $2^{\eta t} \geq \max\{e^{16}, N_{\mathrm{w}}\}$. Then theorem 2.2 yields a wheel $W \subseteq H$ in which every vertex lies in at least

$2^{\rho N/\log^{16} N - 1} \geq 2^{ct}$ distinct Hamiltonian subsets. In the bipartite case W contains a cycle and hence meets both sides, so the vertex can be chosen in either prescribed side. □

5. Proof of the main theorem

Proof of theorem 1.1. Let c, d_1 be supplied by theorem 4.1, and set $\kappa = 1/100$. Choose $d_0 \geq d_1$ so large that every graph considered below has $n \geq 100$, $t \geq 4$, and $ct \geq 4$; this is possible because $n > \overline{d}(G)$ and $t > \overline{d}(G)/5$ by (1).

Suppose that the theorem fails, and choose a counterexample G with the fewest vertices. Write $n = v(G)$, $d = \overline{d}(G)$, $t = c_{1/5}(G)$ and $m = e(G) = dn/2$. Thus $h(G) < \kappa n 2^{ct}$, whereas every graph J with $v(J) < n$ and $\overline{d}(J) \geq d_0$ satisfies

$$h(J) \geq \kappa v(J) 2^{c c_{1/5}(J)}. \tag{4}$$

Set $r = \lceil 2\kappa n \rceil$.

Stage 1. Start with $S = \varnothing$. While $\overline{d}(G - S) \geq d/4$, apply theorem 4.1 to $F = G - S$. Choose a vertex $v \notin S$ contained in at least 2^{ct} Hamiltonian subsets of $G - S$, record those subsets, and add v to S. Families recorded in different iterations are disjoint: the family recorded when v is chosen consists of sets containing v, whereas all later families avoid v.

If this stage has r iterations, then $h(G) \geq r2^{ct} \geq 2\kappa n 2^{ct}$, a contradiction. It therefore stops after fewer than r iterations, with

$$e(G - S) < \frac{d(n - |S|)}{8} \leq \frac{m}{4}. \tag{5}$$

Stage 2. For the current S, let F_S be the spanning bipartite subgraph of G consisting of the edges between S and $V(G) \setminus S$. While $e(F_S) \geq m/4$, we have $\overline{d}(F_S) = 2e(F_S)/n \geq d/4$. Apply the bipartite form of theorem 4.1, prescribing the side $V(G) \setminus S$. Choose a vertex v in that side contained in at least 2^{ct} Hamiltonian subsets of F_S, record those subsets, and add v to S.

Families recorded in different iterations of Stage 2 are disjoint. Indeed, let $S_i \subsetneq S_j$ be the sets before iterations $i < j$, and let v_i be the vertex added at iteration i. If a set A were recorded in both iterations, then a spanning cycle on A in each corresponding bipartite graph would give $|A \cap S_i| = |A|/2 = |A \cap S_j|$. But $v_i \in A \cap (S_j \setminus S_i)$ and $S_i \subseteq S_j$, so $|A \cap S_j| > |A \cap S_i|$, a contradiction.

If Stage 2 has r iterations, its recorded families alone contradict the choice of G. Thus it stops after fewer than r iterations, with

$$e(F_S) < \frac{m}{4}. \tag{6}$$

The quantity $e(G - S)$ can only decrease as S grows, so (5) still holds. Moreover, $|S| < 2r \leq 4\kappa n + 2 \leq 6\kappa n$. Combining (5) and (6), we obtain

$$e(G[S]) = m - e(G - S) - e(F_S) > \frac{m}{2}. \tag{7}$$

Stage 3. By (7), S is nonempty. Write $|S| = n/b$. Since $|S| < 6\kappa n$ and $\kappa = 1/100$, we have $b > 1/(6\kappa) > 16$. Also,

$$\overline{d}(G[S]) = \frac{2e(G[S])}{|S|} > \frac{m}{|S|} = \frac{b}{2}d.$$

Write $\overline{d}(G[S]) = \gamma d$, so $\gamma > b/2$. Let $t_S = c_{1/5}(G[S])$. Every subgraph of $G[S]$ with average degree at least $\overline{d}(G[S])/(5\gamma) = d/5$ is a $(1/5)$-crux candidate in G. Hence $c_{1/(5\gamma)}(G[S]) \geq t$. By theorem 2.4,

$$t \leq c_{1/(5\gamma)}(G[S]) \leq \lceil \frac{t_S - 1}{\gamma} + 1 \rceil.$$

Since t is an integer, this implies $(t_S - 1)/\gamma + 1 > t - 1$, and therefore

$$t_S > \gamma(t - 2) + 1 > \frac{bt}{4}, \tag{8}$$

where the last inequality uses $t \geq 4$ and $\gamma > b/2$.

The graph $G[S]$ has fewer than n vertices and average degree greater than $bd/2 > d \geq d_0$. Applying (4) and then (8),

$$h(G) \geq h(G[S]) \geq \kappa \frac{n}{b} 2^{ct_S} = \kappa n 2^{ct} \frac{2^{c(t_S - t)}}{b}.$$

Because $b > 16$, (8) gives $t_S - t > bt/8$. Since $ct \geq 4$, we have $c(t_S - t) > b/2 \geq \log_2 b$, and hence $2^{c(t_S - t)} > b$. It follows that $h(G) > \kappa n 2^{ct}$, contradicting the choice of G. $\square$

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